

Exposé VIII, continued. Pairings and the first group-scheme cases

2.1.11. The anti-commutativity sign

We continue the construction following Lemma 2.1.10. With the notation used there, the two possible passages from the double torsion class to the derived tensor product differ by the usual sign. The point is that the two standard resolutions used to compute Tor_1 are interchanged by the transposition in the derived tensor product, and this transposition has degree one in each variable.

Thus, in the notation of the preceding page, the diagram

$$\begin{array}{ccc} {}_mP \otimes {}_nQ & \longrightarrow & \mathrm{Tor}_1(P, Q) \\ \downarrow - & & \downarrow \sim \\ {}_nQ \otimes {}_mP & \longrightarrow & \mathrm{Tor}_1(Q, P) \end{array} \quad (2.1.11.1)$$

commutes. Equivalently, if $p \in {}_mP$ and $q \in {}_nQ$, the class obtained from (p, q) and then transposed is the negative of the class obtained from (q, p) . This is the anti-commutativity asserted in Lemma 2.1.10.

A second formulation, useful below, is obtained after applying an extension class. If $u : P \otimes^L Q \rightarrow G[1]$ represents a biextension class, the two edge maps

$${}_mP \otimes {}_nQ \longrightarrow G/(m, n)G$$

which are obtained by first using P, Q , or by first transposing Q, P , differ by the same sign. This is the sign which will make the self-pairing alternating in the familiar special cases.

2.1.12. A useful corollary

Let m, n be positive integers and put $d = (m, n)$. The construction just recalled gives a canonical morphism

$${}_mP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1(P, Q), \quad (2.1.12.1)$$

which factors through the quotient by d -multiples in the expected way. If $m = n$, it is compatible with the anti-commutativity isomorphism under interchange of the two variables.

In diagrammatic form, the square

$$\begin{array}{ccc} {}_mP \otimes {}_nQ & \longrightarrow & \mathrm{Tor}_1(P, Q) \\ \downarrow -\tau & & \downarrow \sim \\ {}_nQ \otimes {}_mP & \longrightarrow & \mathrm{Tor}_1(Q, P) \end{array} \quad (2.1.12.2)$$

commutes. Here τ is the transposition of the two factors. The sign is the only non-trivial point: it is precisely the sign coming from the symmetry constraint in the derived category.

2.1.13. Passage from torsion to Ext

Let now G be another abelian Group. For every abelian Group E , write ${}_nE$ for the kernel of multiplication by n . The short exact sequence

$$0 \longrightarrow {}_nG \longrightarrow G \xrightarrow{n} G \quad (2.1.13.1)$$

and the distinguished triangle attached to multiplication by n give the usual connecting homomorphisms. Combining them with (2.1.12.1) gives, functorially in P, Q, G , a morphism

$$\mathrm{Ext}^1(P \otimes^L Q, G) \longrightarrow \mathrm{Hom}({}_nP \otimes {}_nQ, G/nG). \quad (2.1.13.2)$$

Using the canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^L Q, G),$$

we obtain the pairing attached to a biextension.

If E is a biextension of (P, Q) by G , its class therefore defines, for every n , a biadditive pairing

$$e_{n,E} : {}_nP \times {}_nQ \longrightarrow G/nG. \quad (2.1.13.3)$$

The source text records this construction as the homomorphism (2.1.13.3) and then checks its compatibility with the sign diagrams of 2.1.11 and 2.1.12. The anti-commutativity diagram is

$$\begin{array}{ccc} {}_nP \otimes {}_nQ & \xrightarrow{e_{n,E}} & G/nG \\ \downarrow -\tau & & \parallel \\ {}_nQ \otimes {}_nP & \xrightarrow{e_{n,t_E}} & G/nG. \end{array} \quad (2.1.13.4)$$

Thus the pairing of the transposed biextension is the negative transpose of the original pairing.

2.1.14. The case of a prime number

Let ℓ be a prime. If, for every abelian Group E , we put

$$T_\ell(E) = \varprojlim_r {}_{\ell^r}E, \quad (2.1.14.1)$$

then the preceding pairings are compatible with the transition morphisms and hence give a projective-system pairing. In particular, a biextension E of (P, Q) by G gives

$$T_\ell(P) \otimes T_\ell(Q) \longrightarrow T_\ell(G) \quad (2.1.14.2)$$

whenever the indicated inverse limits are taken in the ambient topos and the standard exactness conditions hold.

More explicitly, the source constructs the homomorphism by means of the inverse system

$${}_{\ell^r}P \otimes {}_{\ell^r}Q \longrightarrow G/\ell^r G \longrightarrow T_\ell(G),$$

and records its compatibility with the systems obtained after passing to ℓ^r -torsion in one variable and to the transition $\ell^{r+1} \mapsto \ell^r$ in the other.

If $G = \mathbb{G}_m$, this becomes a pairing with values in

$$T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1),$$

namely the usual Tate-twist value. The anti-commutativity of 2.1.11 says that, after identifying the two variables when appropriate, the induced form is alternating in the graded sense.

2.2. Explicit description by resolutions

The previous construction can be made completely explicit. Choose a flat resolution of Q , written as a two-term complex in degrees 1, 0:

$$0 \longrightarrow M_1 \xrightarrow{d} M_0 \longrightarrow Q \longrightarrow 0. \quad (2.2.1)$$

Tensoring with P gives a complex whose first homology is $\mathrm{Tor}_1(P, Q)$. Hence a class

$$x \in {}_n P, \quad y \in {}_n Q,$$

represented after a lift $y_0 \in M_0$, gives a cycle in $P \otimes M_1$ after multiplying by n . Applying an extension class representing the biextension gives the value of the pairing.

This is the content of the diagrams displayed in the source around (2.2.3)–(2.2.10). The important points are as follows. The section of $\mathrm{Ext}^1({}_n Q, G)$ obtained from the biextension and a torsion section $x \in {}_n P$ is represented by a homomorphism of extensions

$$\begin{array}{ccccccccc} 0 & \longrightarrow & G & \longrightarrow & E_x & \longrightarrow & Q & \longrightarrow & 0 \\ & & \parallel & & \downarrow & & \downarrow n & & \\ 0 & \longrightarrow & G & \longrightarrow & E_x & \longrightarrow & Q & \longrightarrow & 0, \end{array} \quad (2.2.2)$$

where E_x is the extension of Q by G obtained by restricting the biextension to x . Pulling back along ${}_n Q \rightarrow Q$ and then applying the boundary map yields the pairing

$${}_n P \otimes {}_n Q \longrightarrow G/nG.$$

The source checks that this description agrees with the homological one coming from $P \otimes^L Q$.

2.3. Comparison with the classical construction

The preceding formula is now compared with the elementary construction obtained by choosing local sections of the biextension. Suppose $x \in {}_n P$ and $y \in {}_n Q$. If E is a biextension, choose locally a section s of E above (x, y) . Adding s to itself n times in the first variable gives a section above $(0, y)$, and hence, using the rigidification on the first axis, an element of G . Adding in the second variable gives another element. Their quotient is the value

$$e_{n,E}(x, y) \in G/nG. \quad (2.3.1)$$

The biextension axiom, equivalently the compatibility of the two partial group laws, is exactly what makes this element independent of the section chosen and bilinear in (x, y) .

The source writes this verification in cocycle language. If the biextension is represented by a 1-cocycle and by a factor of automorphy, the difference between the two possible n -fold additions is a 1-cocycle with values in G . Its class is precisely the image of the corresponding Ext-class under the edge homomorphism of 2.1.13. The displayed identities (2.3.10)–(2.3.18) in the source show that the cocycle obtained by the local-section calculation is cohomologous to the one obtained from the homological construction.

The comparison can be summarized by the commutative diagram

$$\begin{array}{ccc} \mathrm{Biext}^1(P, Q; G) & \xrightarrow{\sim} & \mathrm{Ext}^1(P \otimes^L Q, G) \\ \downarrow & & \downarrow \\ \mathrm{Hom}({}_n P \otimes {}_n Q, G/nG) & \xlongequal{\quad} & \mathrm{Hom}({}_n P \otimes {}_n Q, G/nG). \end{array} \quad (2.3.2)$$

The left vertical arrow is the elementary biextension pairing; the right vertical arrow is the boundary map obtained from the derived tensor product.

2.3.19. Deligne's second proof

Deligne's proof recorded in the source gives another verification of the sign in 2.1.11. One resolves P and Q by multiplication by n , compares the two acyclic two-term complexes, and observes that the diagram

$$\begin{array}{ccc} nQ & \longrightarrow & Q \\ \downarrow & \swarrow \text{dashed} & \downarrow \\ Q & \longrightarrow & Q \end{array} \quad (2.3.19.1)$$

anti-commutes in degree one. Passing through the chain of canonical identifications

$$\mathrm{Tor}_1(P, Q) \simeq H_1(P \otimes^L Q) \simeq H_1(Q \otimes^L P) \simeq \mathrm{Tor}_1(Q, P)$$

therefore introduces exactly the minus sign. This gives an independent proof that the diagram of 2.1.11 is anti-commutative.

2.4. Change of topos

Let $f : \mathcal{T}' \rightarrow \mathcal{T}$ be a morphism of topoi. Then pullback gives a canonical homomorphism

$$\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(f^*P, f^*Q; f^*G). \quad (2.4.1)$$

On the homological side this is the usual morphism

$$f^* R\mathrm{Hom}(P \otimes^L Q, G) \rightarrow R\mathrm{Hom}(f^*P \otimes^L f^*Q, f^*G),$$

and the source records the compatibility by a commutative square

$$\begin{array}{ccc} \mathrm{Biext}^1(P, Q; G) & \xrightarrow{\sim} & \mathrm{Ext}^1(P \otimes^L Q, G) \\ f^* \downarrow & & \downarrow f^* \\ \mathrm{Biext}^1(f^*P, f^*Q; f^*G) & \xrightarrow{\sim} & \mathrm{Ext}^1(f^*P \otimes^L f^*Q, f^*G). \end{array} \quad (2.4.2)$$

The analogous diagrams for the pairings of 2.1–2.3 commute as well. This is an immediate consequence of the functoriality of the constructions in the preceding exposé and of the compatibility of derived tensor product with inverse image.

3. Biextensions of group schemes (P, Q) by $\mathbb{G}_m$: generalities

In this section the ambient topos is the fppf topos of a base scheme S , and P, Q are group schemes over S . We apply the preceding formalism in the important case where the coefficient group is the multiplicative group $\mathbb{G}_m$.

3.1. The case of an abelian scheme and its dual

Let A be an abelian scheme over S , and let A' be the dual abelian scheme. The Poincare invertible sheaf on $A \times_S A'$, rigidified along the two zero sections, carries a canonical biextension structure by $\mathbb{G}_m$. Its partial extension structures are the theorem of the square in each variable; their compatibility is the theorem of the cube. This biextension represents the identity morphism of A' , and it is the fundamental example to which the following statements are compared.

3.2. One variable abelian

More generally, if Q is an abelian scheme, then

$$\mathrm{Hom}(G, Q) = 0 \tag{3.2.1}$$

for every smooth affine group G in the cases considered here; consequently, for a suitable group scheme P , biextensions of (P, Q) by $\mathbb{G}_m$ are controlled by morphisms from P to the dual abelian scheme Q' . In the form recorded in the source,

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q'), \tag{3.2.2}$$

whenever the hypotheses ensuring representability and rigidity are satisfied. The inverse construction pulls the Poincare biextension on $Q \times_S Q'$ back along $P \rightarrow Q'$.

If P is also abelian, the symmetric statement gives

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(Q, P') \simeq \mathrm{Hom}(P, Q'), \tag{3.2.3}$$

where the last isomorphism is the usual transposition duality.

3.3. Finite locally free layers and duality

For every integer $n > 0$, the n -torsion subschemes ${}_n P$ and ${}_n Q$ intervene in the pairings of Section 2. In the abelian-scheme case they are finite locally free over S . The pairing attached to the Poincare biextension is the Weil pairing

$${}_n A \times {}_n A' \longrightarrow \mu_n. \tag{3.3.1}$$

Equivalently, the homomorphism ${}_n A \rightarrow ({}_n A')^D$ induced by the biextension is an isomorphism, and likewise after interchanging the two variables. Thus the source recovers the biduality theorem of Barsotti and Cartier from the biextension formalism.

At the end of the present range the source records that a morphism $u : P \rightarrow Q'$ may be interpreted, indifferently, as a biextension of (P, Q) by $\mathbb{G}_m$, or as a morphism $v : Q \rightarrow P'$. In the abelian case the two interpretations are equivalent by the canonical identification $P \simeq P''$ and $Q \simeq Q''$.